# DHAKA AIR QUALITY ANALYSIS

**Exploring Pollution Trends and Weather Relationships**

Study Period: August 2022 – February 2026

Study Area: Dhaka, Bangladesh



## Overview

Air pollution is a significant environmental challenge in Dhaka, where pollutant concentrations vary across seasons and weather conditions. This project investigates air-quality data to identify pollution trends, examine relationships between different pollutants, and explore how meteorological factors are associated with PM2.5 concentrations.

The project combines Python-based data analysis, DuckDB SQL queries, and Power BI visualisation to transform environmental data into meaningful insights.

## Research Objectives

1.Analyse historical PM2.5 concentrations and identify monthly and seasonal pollution patterns.

2.Investigate correlations between PM2.5 and other atmospheric pollutants.

3.Examine the relationship between meteorological variables and PM2.5 concentrations

4.Identify high-pollution episodes and investigate which pollutants remain elevated during these periods.

## Dataset

The analysis uses  air-quality observations and meteorological data covering Dhaka from August 2022 to February 2026.

**Air-quality variables:** PM2.5, PM10, carbon monoxide, nitrogen dioxide, sulphur dioxide, ozone and other air-quality indicators.

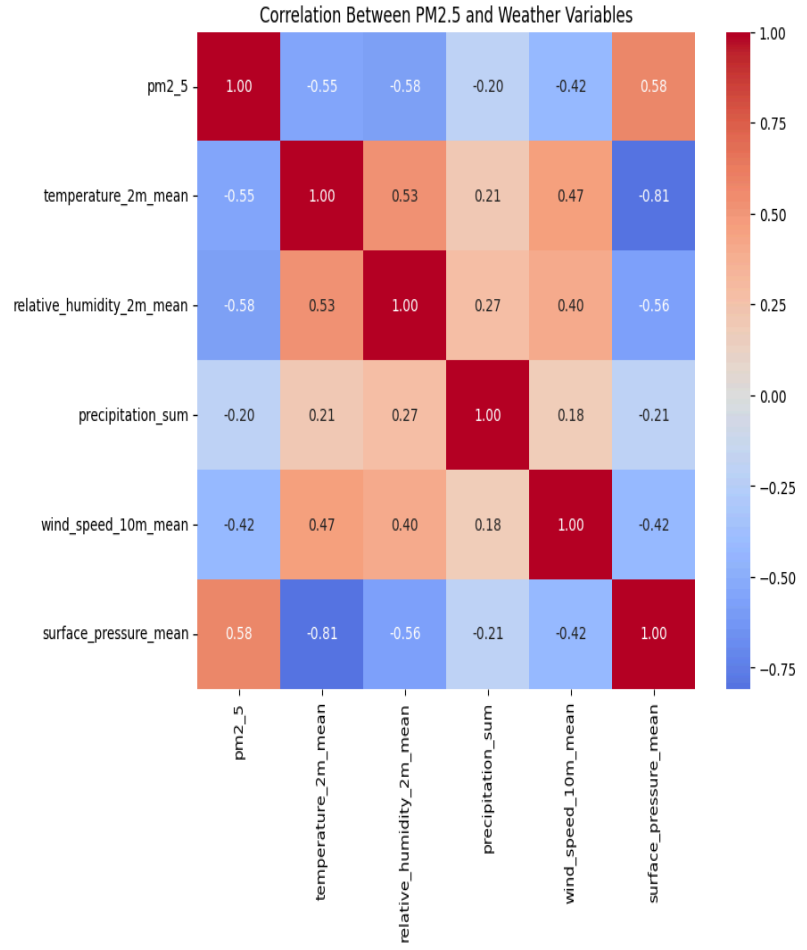
**Meteorological variables:** Mean temperature, relative humidity, precipitation, wind speed and surface pressure.

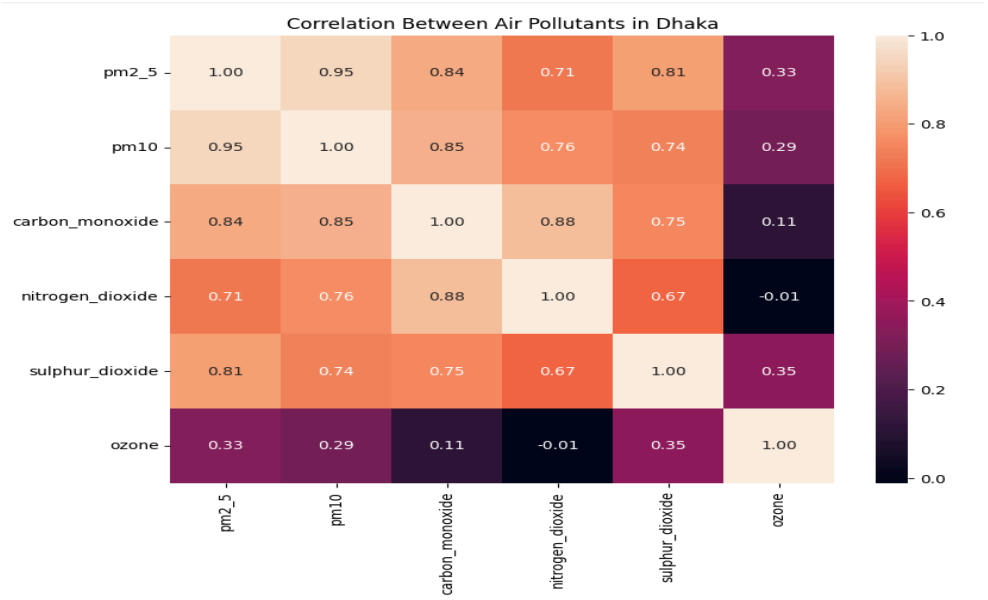
## Key Findings

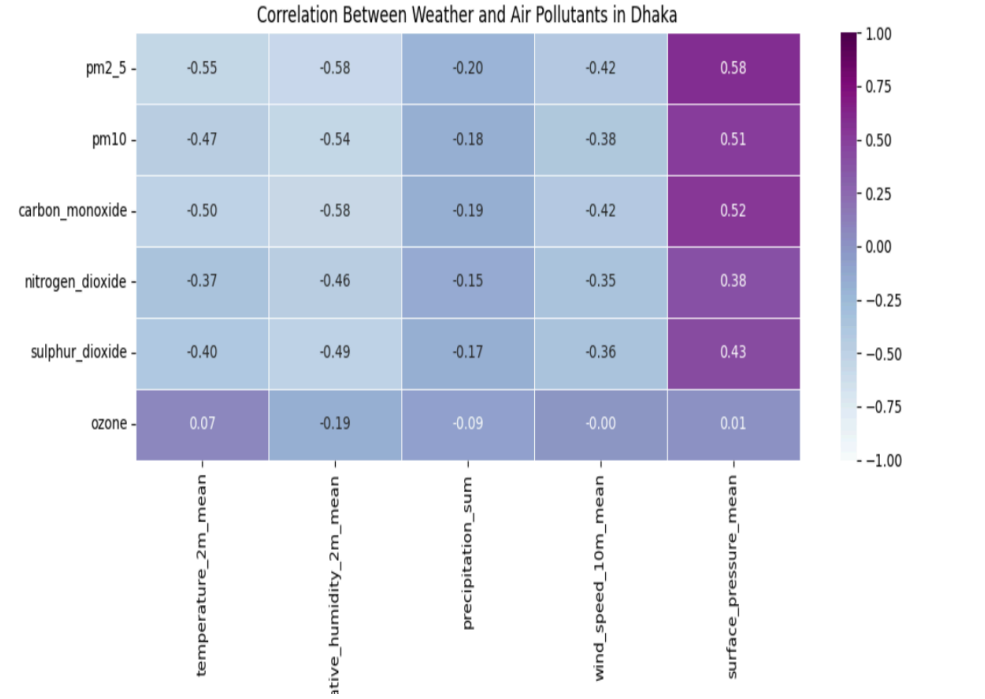
1. Winter recorded the highest average PM2.5 concentration (84.95), while monsoon recorded the lowest (21.90), revealing substantial seasonal variation in Dhaka's air quality.

2. Monthly pollution trends: January had the highest average PM2.5 concentration (98.17), followed by February (80.76) and December (75.22).
3. Pollutant relationships: PM2.5 showed a very strong positive correlation with PM10 (r = 0.947), followed by carbon monoxide (r = 0.835) and nitrogen dioxide (r = 0.713).
4. Weather-related associations: PM2.5 was negatively correlated with temperature (r = −0.621), wind speed (r = −0.550), relative humidity (r = −0.485) and precipitation (r = −0.359).
5. High-pollution patterns: Using a 90th-percentile PM2.5 threshold of approximately 72.61, 324 observations were classified as high-pollution days. Episode-level analysis further examined the duration and pollutant behaviour of sustained high-pollution periods.

## Visual analysis






## Conclusion

This project demonstrates how environmental datasets can be analysed using statistical methods, SQL and data visualisation to investigate urban air-quality patterns. The findings highlight seasonal differences in PM2.5 concentrations, relationships between pollutants and associations between meteorological conditions and air pollution.